

AERO70015/96012 Mathematics 3 Problem Set #2

Problem Sheet 2

2.1 PR2.1

By writing $f(z) = u(x, y) + i v(x, y)$, where $z = x + i y$, find the real functions $u(x, y)$ and $v(x, y)$ in each of the following cases:

- (a) $z^2 + 2z$
- (b) $z/(e^z - 1)$ ($z \neq 0$)
- (c) $\sin z$
- (d) $1/z$ ($z \neq 0$)

$$\begin{aligned} \text{(a)} \quad z^2 + 2z &= (x + iy)^2 + 2(x + iy) \\ &= x^2 + 2ixy - y^2 + 2x + 2iy \\ &= (x^2 - y^2 + 2x) + i(2xy + 2y) \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \frac{z}{e^z - 1} &= \frac{x + iy}{e^{x+iy} - 1} = \frac{x + iy}{e^x e^{iy} - 1} \\ &= \frac{x + iy}{e^x (\cos y + i \sin y) - 1} = \frac{x + iy}{(e^x \cos y - 1) + i e^x \sin y} \\ &= \frac{(x + iy)(e^x \cos y - 1 - i e^x \sin y)}{(e^x \cos y - 1)^2 + (e^x \sin y)^2} \\ &= \frac{x e^x \cos y - x + y e^x \sin y}{e^{2x} (\cos^2 y + \sin^2 y) - 2 e^x \cos y + 1} + i \frac{y e^x \cos y - y - x e^x \sin y}{e^{2x} (\cos^2 y + \sin^2 y) - 2 e^x \cos y + 1} \\ &= \frac{x e^x \cos y - x + y e^x \sin y}{e^{2x} - 2 e^x \cos y + 1} + i \frac{y e^x \cos y - y - x e^x \sin y}{e^{2x} - 2 e^x \cos y + 1} \end{aligned}$$

$$\text{(c)} \quad \sin z = \sin(x + iy) = \sin x \cosh y + i \cos x \sinh y$$

$$\begin{aligned} \text{(d)} \quad \frac{1}{z} &= \frac{1}{x + iy} = \frac{x - iy}{(x + iy)(x - iy)} \\ &= \frac{x}{x^2 + y^2} + i \left(-\frac{y}{x^2 + y^2} \right) \end{aligned}$$

2.2 PR2.2

By using the Cauchy-Riemann equations, show that the function

$$f(z) = \left(1 + \frac{i}{4\pi}\right) \log z$$

is analytic in any region that does not circle the origin.

Show that the equation of any contour on which $u(x, y)$ is constant may be written in the form $r = A e^{\theta/(4\pi)}$, where A is a constant and (r, θ) are polar coordinates. Sketch these contours and also those on which $v(x, y)$ is constant.

What physical situation might these contours describe?

$$\begin{aligned} f(z) &= \left(1 + i \frac{1}{4\pi}\right) \ln(re^{i\theta}) \\ &= \left(1 + i \frac{1}{4\pi}\right) (\ln r + i\theta) \\ &= \left(\ln r - \frac{\theta}{4\pi}\right) + i\left(\theta + \frac{1}{4\pi} \ln r\right) \end{aligned}$$

$$\therefore \frac{\partial u}{\partial r} = \frac{1}{r}$$

$$\frac{\partial u}{\partial \theta} = -\frac{1}{4\pi}$$

$$\frac{\partial v}{\partial r} = \frac{1}{4\pi r}$$

$$\frac{\partial v}{\partial \theta} = 1$$

for $r \neq 0$, where $(x, y) \neq (0, 0)$

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta} \quad \text{and} \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta} \quad \therefore \text{satisfies C-R}$$

$\therefore f(z)$ is analytic despite at origin

if $u = \ln r - \frac{\theta}{4\pi} = \text{constant} = C$

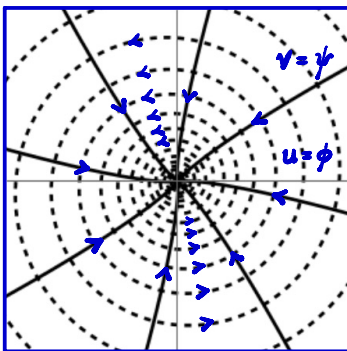
$$\ln r = C + \frac{\theta}{4\pi}$$

$$\therefore r = e^{C + \frac{\theta}{4\pi}} = e^C e^{\frac{\theta}{4\pi}} = A e^{\frac{\theta}{4\pi}}$$

similarly, for v , $r = B e^{-4\pi\theta}$

u might be potential function, ϕ

v might be stream function, ψ



flow with sink at origin

2.3 PR2.3 (imported from corrected file)

Using n to stand for any constant integer, and writing π as pi, determine all the possible values of

- (a) $\log(1 - i)$
- (b) $\arctan(2i)$
- (c) $(1 + i)^{2i}$
- (d) $(1 + i)^{1/2}$ (principal value only)
- (e) 2^{3+2i} (principal value only)

(a) $\ln(1-i)$

for $z = re^{i\theta} = 1 - i$, $r = \sqrt{2}$, $\theta = \arg z = \tan^{-1}(-\frac{1}{1}) = -\frac{1}{4}\pi + 2n\pi$

$$\therefore \ln(1-i) = \ln(re^{i\theta}) = \ln r + i\theta = \ln \sqrt{2} + i(-\frac{1}{4}\pi + 2n\pi)$$

(b) ~~$\tan^{-1}(2i) = \tan^{-1}(2e^{i(\frac{1}{2}\pi + 2n\pi)})$~~

$$z = \tan^{-1}(2i)$$

$$\therefore \tan z = 2i$$

$$\therefore \frac{\sin z}{\cos z} = 2i$$

$$\therefore \frac{e^{iz} - e^{-iz}}{2i} \cdot \frac{2}{e^{iz} + e^{-iz}} = \frac{e^{iz} - e^{-iz}}{i(e^{iz} + e^{-iz})} = i \frac{e^{iz} - e^{-iz}}{e^{iz} + e^{-iz}} = 2i$$

$$\therefore e^{-is} - e^{is} = 2e^{is} + 2e^{-is}$$

$$\therefore e^{is} = -\frac{1}{3}e^{-is}$$

$$\therefore e^{2is} = e^{(2ix - 2y)} = -\frac{1}{3}$$

$$\therefore e^{-2y} (\cos 2x + i \sin 2x) = -\frac{1}{3}$$

$$\therefore e^{-2y} = \frac{1}{3}$$

$$\cos 2x + i \sin 2x = -1$$

$$\therefore y = -\frac{1}{2} \ln \frac{1}{3} = \ln \left(\frac{1}{3}\right)^{-\frac{1}{2}} = \ln \sqrt{3}$$

$$x = \frac{\pi}{2} + n\pi$$

$$\therefore z = \frac{\pi}{2} + n\pi + i \ln \sqrt{3}$$

$$(c) (1+i)^{2i} = e^{2i \ln(1+i)}$$

according to (a), similarly, $\ln(1+i) = \ln \sqrt{2} + i(\frac{1}{4}\pi + 2n\pi)$

$$\therefore (1+i)^{2i} = e^{2i(\ln \sqrt{2} + i(\frac{1}{4}\pi + 2n\pi))} = e^{\ln 2} e^{-\frac{1}{2}\pi - 4n\pi} = e^{-\frac{1}{2}\pi - 4n\pi} (\cos \ln 2 + i \sin \ln 2)$$

$$(d) (1+i)^{\frac{1}{2}} = e^{\frac{1}{2} \ln(1+i)}$$

according to (c), $\ln(1+i) = \ln \sqrt{2} + i(\frac{1}{4}\pi + 2n\pi)$

$$\therefore (1+i)^{\frac{1}{2}} = e^{\frac{1}{2} \ln \sqrt{2}} e^{i(\frac{1}{8}\pi + n\pi)} = 2^{\frac{1}{4}} e^{i\frac{1}{8}\pi} \text{ for principle value} = 2^{\frac{1}{4}} (\cos \frac{1}{8}\pi + i \sin \frac{1}{8}\pi)$$

$$(e) 2^{3+2i} = e^{(3+2i)(\ln 2 + 2n\pi i)}$$

$$= e^{3 \ln 2 - 4n\pi} e^{i(2 \ln 2 + 6n\pi)}$$

$$= 8 e^{i \ln 4} \text{ for principle value}$$

$$= 8 (\cos \ln 4 + i \sin \ln 4)$$

2.4 PR2.4

Evaluate the following using Cauchy's integral formula:

(a)

$$\oint_C \frac{e^z}{z + i\pi/2} dz,$$

where C is the boundary of the square defined by the lines $x = \pm 2$, $y = \pm 2$.

(b)

$$\oint_C \frac{e^z}{z(z+1)} dz,$$

where C is the boundary of the circle $|z - 1| = 3$.

(a) let $f(z) = e^z$, which is entire

$$\therefore \oint_C \frac{f(z)}{z + i\frac{\pi}{2}} dz = 2\pi i f(-i\frac{\pi}{2})$$

the point $z_0 = -i\frac{\pi}{2}$ is inside the path of C

$$\therefore \oint_C \frac{f(z)}{z + i\frac{\pi}{2}} dz = 2\pi i e^{-i\frac{\pi}{2}} = 2\pi i [\cos(\frac{\pi}{2}) - i\sin(\frac{\pi}{2})] = 2\pi i \cdot (-i) = 2\pi$$

$$(b) \oint_C \frac{e^z}{z(z+1)} dz = \oint_C \frac{e^z}{z} - \frac{e^z}{z+1} dz = \oint_C \frac{e^z}{z} dz - \oint_C \frac{e^z}{z+1} dz$$

let $f(z) = e^z$, which is entire

the point $(0,0)$ and $(-1,0)$ is inside the path of C

$$\therefore \oint_C \frac{e^z}{z(z+1)} dz = 2\pi i e^0 - 2\pi i e^{-1} = 2\pi (1 - e^{-1})i$$

2.5 PR2.5

Evaluate the following integrals, where C is the boundary of the circle $|z| = 2$:

(a)

$$\oint_C \frac{\cos z}{z^n} dz, \quad n = 1, 2, \dots$$

(b)

$$\oint_C z^n (1-z)^m dz, \quad m = 0, 1, 2, \dots; \quad n = 0, \pm 1, \pm 2, \dots$$

(c)

$$\oint_C \frac{z^3 + 5}{z - i} dz$$

(a) let $f(z) = \cos z$, which is entire

the point $(0,0)$ is inside the path of C

$$\begin{aligned} \therefore \oint_C \frac{\cos z}{z^n} dz &= \frac{2\pi i}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} (\cos z) \Big|_{z=0} \\ &= \begin{cases} 0, & n = 2, 4, 6, \dots \\ \frac{2\pi i}{(n-1)!}, & n = 1, 5, 9, \dots \\ -\frac{2\pi i}{(n-1)!}, & n = 3, 7, 11, \dots \end{cases} \\ &= \begin{cases} 0, & n \text{ even} \\ (-1)^{\frac{n-1}{2}} \frac{2\pi i}{(n-1)!}, & n \text{ odd} \end{cases} \end{aligned}$$

(b) $\oint_C z^n (1-z)^m dz$

if $n \geq 0$, $z^n (1-z)^m$ is analytic inside $|z| = 2$

therefore, according to Cauchy's Integral Theorem, $\oint_C z^n (1-z)^m dz = 0$

if $n < 0$, let $f(z) = (1-z)^m$, the point $(0,0)$ is inside the path of C

$$\therefore \oint_C z^n (1-z)^m dz = \oint_C \frac{(1-z)^m}{z^{-n}} dz = \frac{2\pi i}{(-n-1)!} \frac{d^{-n-1}}{dz^{-n-1}} (1-z)^m \Big|_{z=0}$$

where $\frac{d^r}{dz^r} (1-z)^m = (-1)^r m(m-1)(m-2) \dots [m-(r-1)] (1-z)^{m-r} = (-1)^r \frac{m!}{(m-r)!} (1-z)^{m-r}$ for $m \geq r$

$$\therefore \oint_C z^n (1-z)^m dz = \begin{cases} 0, & m < -n-1 \\ \frac{2\pi i}{(-n-1)!} (-1)^{-n-1} \frac{m!}{(m+n+1)!}, & m \geq -n-1 \end{cases}$$

(c) let $f(z) = z^2 + 5$, which is entire,

the point $(0,0)$ is inside the path of C ,

$$\oint_C \frac{z^2+5}{z-i} = 2\pi i (i^2+5) = 2\pi i (-1+5) = 2\pi i \cdot 4 = 8\pi i$$

2.6 PR2.6

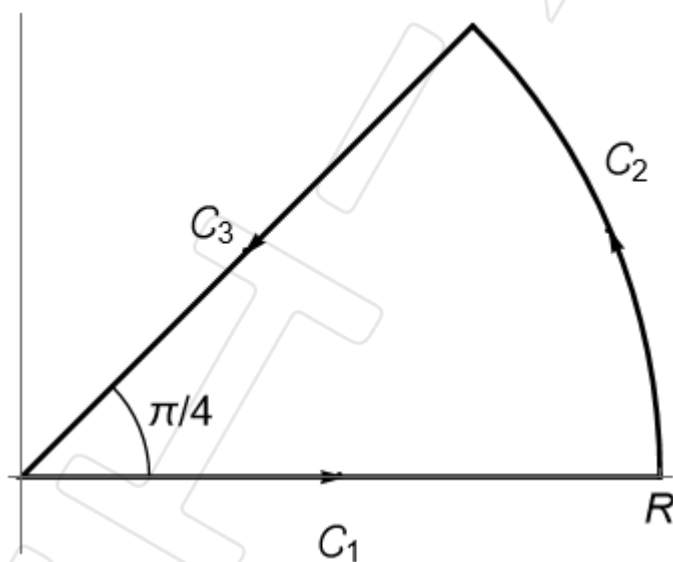
(a)



By considering the integral of $e^{-(z+i)^2}$ along the contour in the figure, and letting R tend to infinity, calculate

$$\int_{-\infty}^{\infty} e^{-x^2-2ix} dx,$$

where you are given that $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$.



(b)

By considering the integral of e^{iz^2} along the contour in the figure, and letting R tend to infinity, calculate

$$\int_{-\infty}^{\infty} e^{ix^2} dx.$$

You may assume that

$$\int_0^{\pi/4} R e^{-R^2 \sin(2\theta)} d\theta$$

tends to zero as R tends to infinity.

$$(a) \oint_C e^{-(z+i)^2} dz = \oint_C e^{-z^2 - 2iz + i} dz = e \oint_C e^{-z^2 - 2iz} dz$$

Also,

$$\begin{aligned} \oint_C e^{-(z+i)^2} dz &= \int_{C_1} e^{-(z+i)^2} dz \Big|_{z=x-i} + \int_{C_2} e^{-(z+i)^2} dz \Big|_{z=x} + \int_{C_3} e^{-(z+i)^2} dz \Big|_{z=R+iy} + \int_{C_4} e^{-(z+i)^2} dz \Big|_{z=-R+iy} \\ &= \int_{-\infty}^{\infty} e^{-x^2} dx + \int_{-\infty}^{\infty} e^{-(x+i)^2} dx + \int_{-i}^i e^{-\infty} dy + \int_0^{-i} e^{-\infty} dy \\ &= \sqrt{\pi} - e \int_{-\infty}^{\infty} e^{-x^2 - 2ix} dx + 0 - 0 \end{aligned}$$

According to Cauchy's Integral Theorem, since $e^{-(z+i)^2}$ is entire, $\oint_C e^{-(z+i)^2} dz = 0$

$$\therefore \int_{-\infty}^{\infty} e^{-x^2 - 2ix} dx = \frac{\sqrt{\pi}}{e}$$

(b) According to Cauchy's Integral Theorem, since e^{iz^2} is entire

$$\oint_C e^{iz^2} dz = 0$$

Also,

$$\begin{aligned} \oint_C e^{iz^2} dz &= \int_{C_1} e^{iz^2} dz \Big|_{z=x} + \int_{C_2} e^{iz^2} dz \Big|_{z=Re^{i\theta}} + \int_{C_3} e^{iz^2} dz \Big|_{z=te^{i\frac{\pi}{4}}} \\ &= \int_0^R e^{ix^2} dx + \int_0^{\frac{\pi}{4}} e^{iR^2(\cos 2\theta + i\sin 2\theta)} iRe^{i\theta} d\theta + \int_{\frac{\pi}{4}}^0 e^{-t^2} e^{it^{\frac{3}{2}}} dt \\ &= \int_0^R e^{ix^2} dx + \int_0^{\frac{\pi}{4}} iRe^{iR^2\cos 2\theta - R^2\sin 2\theta} (\cos \theta + i\sin \theta) d\theta - \int_0^R e^{-t^2} e^{it^{\frac{3}{2}}} dt \end{aligned}$$

$$\begin{aligned} \text{where } \left| \int_0^{\frac{\pi}{4}} iRe^{iR^2\cos 2\theta - R^2\sin 2\theta} (\cos \theta + i\sin \theta) d\theta \right| &\leq |i| \int_0^{\frac{\pi}{4}} |R| \underbrace{e^{iR^2\cos 2\theta}}_{=1} \underbrace{e^{-R^2\sin 2\theta}}_{=1} \underbrace{(\cos \theta + i\sin \theta)}_{=1} d\theta \\ &= i \int_0^{\frac{\pi}{4}} Re^{-R^2\sin 2\theta} d\theta \rightarrow 0 \text{ as } R \rightarrow \infty \end{aligned}$$

$$\therefore \int_0^{\frac{\pi}{4}} iRe^{iR^2\cos 2\theta - R^2\sin 2\theta} (\cos \theta + i\sin \theta) d\theta \rightarrow 0 \text{ as } R \rightarrow \infty$$

$$\text{and } \int_0^R e^{-t} e^{i\frac{\pi}{2}} dt = e^{i\frac{\pi}{2}} \int_0^R e^{-t} dt = e^{i\frac{\pi}{2}} \left(\frac{1}{2} \sqrt{x} \right) \text{ as } R \rightarrow \infty$$

$$\therefore \int_0^R e^{ix} dx = \sqrt{x} e^{i\frac{\pi}{2}}$$

2.7 Past Exam Question Part, 2017

(a) The complex function $f(z)$ is given by

$$f(z) = u(x, y) + i v(x, y),$$

where x and y are, respectively, the real and imaginary parts of z , and u and v are differentiable functions from $\mathbb{R}^2$ to $\mathbb{R}$.

By considering the two limits as $\delta z \rightarrow 0$ of

$$\frac{f(z_0 + \delta z) - f(z_0)}{\delta z}$$

as the point $z_0 = x_0 + i y_0$ is approached along paths parallel to the real axis and the imaginary axis, prove that if $f(z)$ is complex-differentiable at z_0 , then at (x_0, y_0) u and v satisfy the Cauchy-Riemann equations,

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y}, \\ \frac{\partial v}{\partial x} &= -\frac{\partial u}{\partial y}. \end{aligned}$$

(b) Find, in terms of the real variables x and y , the real and imaginary parts, u and v respectively, of $(x + i y)^3$, and verify that they satisfy the Cauchy-Riemann equations everywhere.

By considering the partial derivatives $\partial u / \partial x$ and $\partial v / \partial x$, show that the complex derivative of z^3 is $3 z^2$.

(a) when along the path parallel to the real axis,

$$\frac{\partial f}{\partial z} = \frac{u(x_0 + \delta x, y_0) - u(x_0, y_0) + i[v(x_0 + \delta x, y_0) - v(x_0, y_0)]}{\delta x} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

when along the path parallel to the imaginary axis,

$$\frac{\partial f}{\partial z} = \frac{u(x_0, y_0 + \delta y) - u(x_0, y_0) + i[v(x_0, y_0 + \delta y) - v(x_0, y_0)]}{i \delta y} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

equate both equation. therefore, get

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

$$(b) (x + iy)^3 = (x^3 + 2ixy - y^3)(x + iy) = x^3 + 2ix^2y - xy^3 + ix^2y - 2xy^3 - iy^3$$

$$= (x^3 - 3xy^2) + i(2x^2y + x^2y - y^3)$$

$$\therefore u = x^3 - 3xy^2$$

$$v = 2x^2y + x^2y - y^3$$

$$\therefore \frac{\partial u}{\partial x} = 3x^2 - 3y^2$$

$$\frac{\partial u}{\partial y} = -6xy$$

$$\frac{\partial v}{\partial x} = 4xy + 2xy = 6xy$$

$$\frac{\partial v}{\partial y} = 2x^2 + x^2 - 3y^2 = 3x^2 - 3y^2$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$\therefore$ satisfied C-R equation, $(x+iy)^3$ is entire

$$\therefore \frac{\partial f}{\partial z} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$= 3x^2 - 3y^2 + 6ixy$$

$$= 3(x+iy)^2$$